

What is in the Universe?

Setting the scene

When you look into the night sky you see lots of different bright objects. They are so far away they can look the same. This activity helps you understand your place in the Universe.

Aims

In this activity you will:

- name different objects in the night sky
- describe different objects in the night sky
- describe your place in the Universe.

You will be **working scientifically** to:

- understand that scientific methods and theories develop as scientists modify earlier explanations to take account of new evidence and ideas.

Task

Read the text carefully, and answer the questions that follow.

When we send a letter abroad we include the house name or number, street, town, and country. The Universe includes everything that exists. If we could send a message to another place in the Universe we would need more information in the address.

We live on a planet called Earth. Planets are smaller than stars and travel around them. Earth is our planet and it orbits the Sun. Moons are smaller than planets and travel around them. Earth has one moon. A Solar System includes a star, its planets, and their moons.

All stars give out heat and light. We can see the planets that orbit the Sun because they reflect the Sun's light. The star closest to the Sun is called Proxima Centauri.

Billions of stars grouped together form a galaxy. Our galaxy is called the Milky Way galaxy, and it is shaped in a spiral. The Sun is in one of the galaxy's arms. The closest galaxy to the Milky Way is the Andromeda galaxy. There are billions of other galaxies in the Universe.

Questions

1 State the objects in the Universe that give out their own light.

2 State the objects in the Universe that reflect light from stars.

3 List the objects found in the Universe in order of size, starting from the smallest.

4 Describe the different types of objects found in the Universe.

5 Write down the address of Earth in as much detail as possible.

6 Explain which parts of the Universe are in the Solar System and in a galaxy.

Extension

1 It is hard to imagine how big a billion is. A billion is 1 000 000 000 (or one thousand million). There are 1 000 millimetres in 1 metre and 1 000 metres in 1 kilometre. How many millimetres are in 1 kilometre?

2 How many kilometres would make one billion millimetres?

3 The distance from the tip of Scotland to the bottom of England is 1 340 km. Is this longer than one billion millimetres?
